Networking Basics: Wi-Fi

What is Wi-Fi?

Wi-Fi is a wireless networking technology that allows devices such as laptops, smartphones, tablets, and smart TVs to connect to the internet or communicate with each other without using cables.

Wi-Fi uses **radio waves** to transmit data between devices and a **wireless router**.

How Wi-Fi Works

1. Your Internet Service Provider (ISP) provides an internet connection.
2. The **router** receives the internet connection.
3. The router broadcasts a **Wi-Fi signal**.
4. Devices connect to the Wi-Fi network using the network name (SSID) and password.
5. Data is exchanged wirelessly between the device and the router.

Main Components of a Wi-Fi Network

- **Modem** – Connects your home or office to the internet from the ISP.
- **Wireless Router** – Creates the Wi-Fi network and shares the internet.
- **Access Point (AP)** – Extends or provides wireless coverage in larger areas.
- **Client Devices** – Phones, laptops, tablets, printers, smart TVs, etc.

Common Wi-Fi Standards

Standard	Frequency	Maximum Speed
Wi-Fi 4 (802.11n)	2.4 GHz & 5 GHz	Up to 600 Mbps
Wi-Fi 5 (802.11ac)	5 GHz	Up to 3.5 Gbps
Wi-Fi 6 (802.11ax)	2.4 GHz & 5 GHz	Up to 9.6 Gbps
Wi-Fi 6E	2.4, 5 & 6 GHz	Faster with less interference
Wi-Fi 7	2.4, 5 & 6 GHz	Very high speeds and lower latency

Wi-Fi Frequency Bands

2.4 GHz

Advantages

- Longer range
- Better wall penetration

Disadvantages

- Slower speed
- More interference from other devices

5 GHz

Advantages

- Faster speeds
- Less interference

Disadvantages

- Shorter range
- Doesn't pass through walls as well

6 GHz (Wi-Fi 6E/7)

Advantages

- Fastest speeds
- Very low interference

Disadvantages

- Shortest range
- Requires compatible devices

Wi-Fi Security

To protect your network, use:

- **WPA2** – Secure and widely used.
- **WPA3** – Latest and most secure.
- Avoid using **WEP**, as it is outdated and insecure.

Common Wi-Fi Terms

- **SSID** – The name of the Wi-Fi network.
- **Password** – Used to authenticate users before connecting.
- **Signal Strength** – Indicates how strong the Wi-Fi signal is.
- **Bandwidth** – The amount of data that can be transmitted.
- **Latency (Ping)** – The delay in data transmission.

Advantages of Wi-Fi

- No cables required.
- Easy to install and use.
- Supports multiple devices.
- Enables mobility within the coverage area.
- Cost-effective for homes and offices.

Disadvantages of Wi-Fi

- Signal strength decreases with distance.
- Walls and obstacles can weaken the signal.
- Wireless interference may reduce performance.
- Generally less secure than wired connections if not configured properly.

Common Wi-Fi Problems

- Weak signal
- Slow internet speed
- Frequent disconnections
- Incorrect password
- Router overheating
- Too many connected devices
- Interference from nearby networks

Tips to Improve Wi-Fi Performance

- Place the router in a central, open location.
- Keep the router away from metal objects and thick walls.
- Use the 5 GHz or 6 GHz band when supported.
- Update the router's firmware regularly.
- Secure the network with WPA2 or WPA3.
- Restart the router occasionally.

Interview Questions

1. What is Wi-Fi?
2. What is the difference between a modem and a router?
3. What is an SSID?
4. What are the differences between 2.4 GHz and 5 GHz?
5. What is WPA3?
6. What is an access point?
7. What causes Wi-Fi interference?
8. How can you improve Wi-Fi signal strength?
9. What is the difference between Wi-Fi 5 and Wi-Fi 6?
10. Why should WEP not be used?